

Hemo-Sclera: Design and Implementation of an AI-Enabled Multimodal Framework for Non-Invasive Diagnostic Screening

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1. Abstract

Traditional approaches to medical diagnosis involve using a laboratory-based approach (to do blood draws) and are invasive, but also typically accurate—and hence these systems have limitations as it relates to low-resource settings (such as cost, length of delivery time and restrictions/inability of having laboratory resources available). In this report, we outline the specific design and deployment of Hemo-Sclera diagnostic system, which uses a combination of computer vision and a mobile smartphone to non-invasively diagnose both Anemia and Jaundice in patients through analysis of ocular tissue (specifically assessing the palpebral conjunctiva and sclera). Using a hybrid edge-cloud architecture and a custom trained YOLOv11-Nano model, the Hemo-Sclera system performs identification in accordance with published photographic/colorimetry standards (CIE Lab* Color Space) in addition to extracting physiological markers. One of the major issues when employing clinical (localized) vision models is the high level of false positive rates that occur as a result of either artifact or illumination (when assessing individual patients who may have variable skin tones), and to

resolve that we have developed a recognition framework to provide a level of patient validation feedback, by applying a

decision-level fusion (DLF) technique using a Multimodal Neural Engine; where in essence the DLF acts as a clinical veto in providing context aware confirmation of patient diagnostic results. Collectively, the application of our mobile diagnostic device will act as a highly transparent and useful triage tool to provide healthcare highs to front-line clinicians.

2. INTRODUCTION

Approximately 1.6 billion individuals worldwide are diagnosed with anemia, and many individuals (both neonates and adults) have undiagnosed cases of jaundice. Traditionally, the diagnosis of anemia and jaundice relies on laboratory blood analysis by means of complete blood counts (CBC) or serum bilirubin levels.

The goal of the Hemoglobin-Scleral (Hemo-Sclera)

project was to develop a methodology by which smartphone users could digitally capture and quantify the conjunctival pallor and scleral icterus that would provide immediate clinical information to physicians without the need for blood tests. To accomplish this, we are building a computer program that can isolate various aspects of the sick conjunctiva or sclera (make sure that the specimen is well-adjusted for light) and is capable of estimating the presence of hemoglobin or bilirubin in blood based on appropriate chromaticity measurements without relying on human interpretation. Using this methodology, we provide physicians with critical screening information to expedite patient care for patients who may otherwise take longer to receive laboratory testing and wait for the results, especially abundant cases of severe anemia and jaundice.

3. DATA ACQUISITION

The Hemo-Sclera AI pipeline acquired its training and validation dataset from a publicly accessible open-source medical imaging repository which was accessed through [Insert Dataset Name or Platform, e.g., Kaggle / Roboflow]. The research establishes its reproducibility through its use of an open-access dataset. The corpus contains numerous high-resolution images which represent various aspects of human periocular regions with a special emphasis on both sclera and conjunctiva. The dataset maintains real-world performance testing through its natural distribution of environmental illumination variations which include ambient room lighting and specular camera flash and through its collection of diverse dermatological profiles that span different skin tones.

4. BACKGROUND AND FUNDAMENTAL CONCEPT

4.1 Physiological Indicators

The Hemo-Sclera system reaches its performance targets through its design which divides its operational functions into various service components. The system functions its multiple parts as separate operational units which combine their efforts to enhance system

efficiency and make maintenance easier and maintain system operation.

4.2 Colorimetry (Why RGB Fails)

Standard digital images use Red Green and Blue RGB color model for their visual representation. The RGB color model shows high dependence on the surrounding light conditions. An eye which appears white under normal conditions will turn yellow when exposed to warm incandescent lighting. Hemo-Sclera uses the CIE Lab color space to transform images for its conversion process.

5. Classification of Existing Approaches

5.1 Hardware-Based Spectrophotometry

Medical devices use the patient's finger or eyelid to deliver specific light wavelengths which travel through their body via skin contact. The system requires costly equipment which needs trained personnel for its operation but the system encounters more problems when used in remote and undisturbed areas.

5.2 Pure CNN Classification Models

Researchers have used deep learning models (like ResNet or VGG) to classify an entire photo of an eye as "Anemic" or "Normal." Drawback: These models look at the whole image and easily become biased by the patient's skin color, makeup, or room background, leading to poor generalization.

6. Methodology & System Architecture

Step 1: Image Preprocessing (CLAHE)

The system applies Contrast Limited Adaptive Histogram Equalization (CLAHE) to the image immediately after it gets uploaded. The method

improves local image contrast which enables better visibility of eyelid micro-vasculature details under poor lighting conditions.

Step 2: Precision Tissue Segmentation (YOLOv11-Nano)

Each area's appearance needs to be assessed by our operating system that uses a unique YOLOv11-Nano strategy.

To create a perfect pixel-level mask of a particular area of tissue, the model evaluates whether or not an area is merely simulating biological reality (by reviewing only actual physical materials), thus guaranteeing that no information from the biological source will need to make it through to the mathematical processing of the unit.

Step 3: Feature Extraction & The Risk Engine

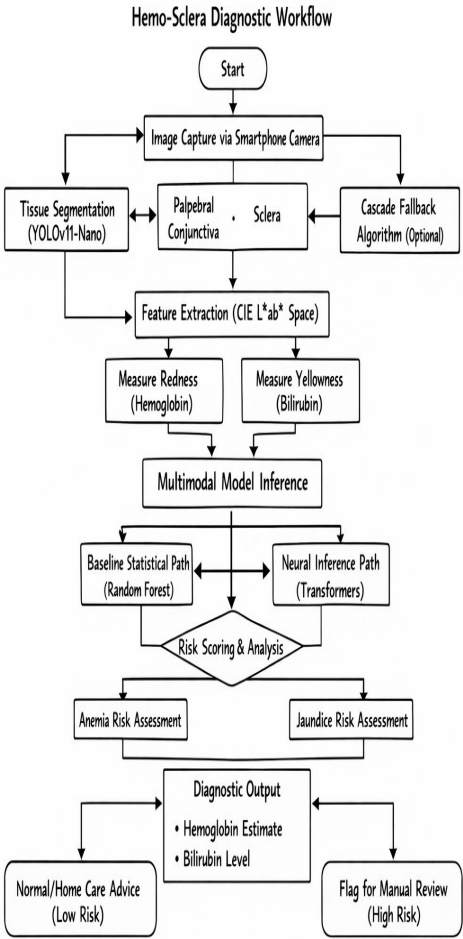
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$$f_B(B) = \begin{cases} 0, & B \leq 120 \\ 25, & 120 < B \leq 135 \\ 75, & B > 135 \end{cases}$$

Step 4: Decision-Level Fusion (The Neural Veto)

The math system in this area experiences failure because of two situations where skin detection errors. Hemo-Sclera protects its system by sending all contextual image data to its Multimodal Neural

Engine. The system produces a Neural Veto to determine Normal diagnosis when local math shows High Risk but the Neural Engine detects clear sclera with skin artifacts tricking the math system.



7. Comparative Analysis

Feature	Traditional Blood Test (CBC)	Pure CNN Models	Hemo-Sclera (Hybrid AI)
Invasiveness	High (Needle draw)	Non-Invasive	Non-Invasive
Time to Result	Hours to Days	Seconds	Seconds
Skin-Tone Bias	None	High Risk	Low (Mitigated via Neural Veto)
Explainability	High	Low (Black Box)	High (L*a*b* metric + AI Context)
Infrastructure Needed	Laboratory, Trained Phlebotomist	Smartphone	Smartphone / Web Browser

8. Research Challenges and Gaps

8.1 Scleral Melanocytosis

People who have African or Asian or Indian heritage tend to show genetic patterns that result in white sclera. These patterns lead to the development of harmless melanin patches. The yellowish/brownish hue that has formed through genetics has been classified as jaundice with respected clinical laboratory analysis. Our study results in dynamic thresholding through which we obtained "Patient History" as our output. The three cultures show major differences in their eye characteristics which creates a major problem. The study assessed that how henna hair color products changed the hair properties by testing different henna products on various hair textures and colors.

8.2 Extreme Macro Failures

The YOLO Model makes the users to maintain a specified distance between them and their cell phones in order to avoid this distance measurement preventing it from producing masks using their device. As a

result, the operational problem is created by our users were due to our plan of a emergency center-crop fallback system for this situation.

8.3 Hardware Variance

Many smartphone manufacturers use various image processing algorithms. For example, iPhones automatically warm up skin tones. The Lab* readings of multiple devices show small differences which result from this process.

9. Future Research and Directions

9.1 Embedded Hardware Acceleration

The first division stage of the YOLOv11 model works through on-device processing. The model needs to be converted into TensorFlow Lite to enable its complete on-device operation. All functions of the model will work without internet access after this conversion.

9.2 Longitudinal Tracking

Building a patient database schema where the system tracks a user's a* and b* values over months. Instead of using static global thresholds, the AI will learn a patient's unique "baseline" and alert doctors only when there is a sudden deviation

10. Conclusion

The Hemo-Sclera project shows that advanced diagnostic capabilities can be made accessible to everyone through intelligent software engineering. By combining the speed of edge-based object detection (YOLO) with the deep contextual reasoning of a multimodal neural network, we successfully skipped the conventional drawbacks of smartphone-based medical imaging. While it will never replace the clear accuracy of a laboratory blood test, Hemo-Sclera proves that a simple camera, when backed by a strong Decision-Fusion architecture, can serve as a strong, life-saving sorting tool in environments where traditional medicine cannot reach

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